Fiscal Year:	FY 2009	**Task Last Updated:**	FY 04/05/2009
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	05/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	Expedition 19/20 NOTE: End date is now 5/31/2012 (from 5/31/2010) per NSSC information and F. Szofran/MSFC (10/2009) NOTE: Changed to FLIGHT project, per F. Szofran/MSFC (10/2009) NOTE: End date changed to 5/31/2010 (from 5/31/2009) per PI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Herlach, Dieter (Inst. fur Raumsimulation) Holland-Moritz, Dirk (Inst. fur Raumsimulation)		
Grant/Contract No.:	NNX07AK27G		
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Performance Goal Text:			

Task Description:	In 1724, Fahrenheit first observed that liquid water could be cooled by a significant amount below its freezing temperature before crystalline ice forms. In 1952, David Turnbull showed that this ability to supercool was a common property of liquids, occurring even in metallic liquids. His experiments demonstrated that a barrier exists to the initial formation of the crystal from the liquid, a process called nucleation. This process is the starting point of not only crystallization, but of most phase transitions of interest to the physical and materials sciences. It is becoming increasingly clear that it plays a fundamental role in many transformations in biological and medical systems as well. The attainment of an improved ability to control nucleation is of central importance for the development of new materials for advanced applications. In almost all cases the classical theory of nucleation, first developed in the late 19th century, is used to analyze and predict nucleation results. There are, however, serious problems with aspects of that theory in significant regimes of its application. Further, it does not properly describe nucleation when the concentrations of the initial and final phases are different. The task of the continuing ground-based research on this contract is the investigation of the icosahedral ordering and associated thermophysical properties in supercooled liquids, primarily of transition metals and their alloys. Measurements are being made of the structure, specific heat, density, viscosity, surface tension and emissivity in the liquids. These data, when combined with the nucleation data, will lead to a better understanding of the complicated nucleation processes that occur in supercooled liquids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The BESL technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions. The discovery of possible new phase transitions in the liquid will lead to a better understanding of bulk metallic glass formation.
Task Progress:	Research funded in part by NASA under grant NNX07AK27G is focused on measurements of the thermophysical and structural properties of equilibrium and supercooled liquids that have relevance to the original QUASI ISS contract. Over the course of this contract, we have made significant improvements in the Beamline Electrostatic Levitation (BESL) technique, decreasing the background scattering by moving the beam stop just beyond the exit Be window (in previous studies it was placed at the detector), and placing a second collimator inside the entrance Be window. The new location for the beam stop eliminated air scattering of the primary beam, which was the major contributing factor to the background. We have developed computer algorithms to correct for more minor contributions to the background that arise from secondary scattering of the sample-diffracted intensity by the Be window and the air path between the window and the detector. With these improvements, we have obtained the most quantitative x-ray scattering data to date for any equilibrium or nonequilibrium liquid. We have developed methods for accurately measuring the liquid specific heats and in collaboration with Bob Hyers, University of Massachusetts, have obtained density and surface tension data during supercooling. Atomic structures were obtained from the scattering data using a Reverse Monte Carlo method. As a check on this method, collaborations have been developed with theoretical physicists (M. Widom, Carnegie Mellon, S. G. Hao, Iowa State University, and Z. Nussinov, Washington University) to obtain structures from molecular dynamics calculations using ab-initio and effective potentials. We have characterized all of these structures in terms of Voronoi polyhedra and bond orientational order (BOO), and Honeycutt-Anderson (HA) parameters. We have extended the HA analysis to include chemical ordering. This past year has been spent in continuing the analysis of the data obtained in the BESL run at the Advanced Photon Source in August 2007, which was funded in part by this contract. Some key results that have been obtained and published over the course of this contract are listed below. 1. Most Ti and Zr-based liquids show the development of local icosahedral order with supercooling. 2. The local icosahedral order changes across the 3d transition metal liquids, becoming more distorted in the early transition metals (such as Ti) and much less distorted in the later transition metals (such as Ni). This suggests a competition between maximizing the local packing density, which favors icosahedral ordering, and satisfying the angular dependence of the 3d bonding potential. 3. Preliminary studies indicate that the addition of oxygen to Ti-Fe-Si liquids increases the amount of icosahedral ordering, although it appears to be distorted. This explains the formation of high-order rational crystal approximants. 4. The local icosahedral order in the liquid couples to the nucleation barrier, lowering the barrier for the formation of the icosahedral quasicrystal (i-phase) and increasing the barrier for the nucleation of crystal phases. This explains why the i-phase is frequently the primary crystallizing phase in Zr-based glasses and why glass formation is so easy in those systems. 5. The icosahedral order increases dramatically on cooling from the melting temperature through the glass transition in liquids that form Zr-based bulk metallic glasses. The measured temperature dependence of the ordering is in good agreement with predictions from geometrical-frustration-based models of the glass transition, providing the strongest experimental support to date for that model. In addition to the key results listed above, which have already been published, we are completing studies of other liquids and preparing the publication of results. Some of the key results are discussed below. Influence of Hf on the liquid structure and formation of non-periodic solid phases in Ti45Zr(38-x)HfxNi17 alloys - While Hf and Zr are nearly identical chemically, we have found that they behave very differently when included together in some alloys. In studies that we made several years ago, which were supported under an NSF grant (DMR_0307410), we noted an abrupt change in the crystal phases that form in as-cast Ti45Zr(38-x)HfxNi17 alloys. The C14 Laves phase (P63/mmc) forms at low Hf concentrations and a large unit cell FCC Ti2Ni-type (cF96) phase forms for high Hf. Phase formation in the rapidly-quenched liquids of these alloys also show an abrupt change near the middle of the Hf/Zr range, with the icosahedral quasicrystal forming for low Hf and a 3/2 crystal rational approximant forming at high Hf. Our studies of supercooled Ti45Zr(38-x)HfxNi17 liquids have revealed several unexpected features. The lattice of the high temperature solid solution phase, ß(Ti/Zr/Hf/Ni), was found to contract with increasing Hf concentration over the composition range studied (up to 38% Hf and 0% Zr). This is surprising give that Hf and Zr are often assumed to have the same size. The liquid atomic volume initially increased with Hf substitution, consistent with a

contracting distance, but went through a maximum value near the concentration boundary where the crystal phases that formed abruptly changed. The thermal expansion coefficient initially decreased with Hf substitution, but went through a minimum at the same alloy concentration. These changes in the physical properties and structures of the liquids near the boundary where we see a radical change in the ordered phases that form indicates again that the liquid structure plays a critical role in phase formation and cannot be ignored.

Ordering in supercooled liquid Cu(100-x)Zrx bulk metallic glass (BMG)-forming alloys Cu-Zr – Cu-Zr alloys have long been known to be easy glass formers. Recently, they were shown to also be members of the growing class of bulk glass forming alloys (BMGs). Since they are one of the few binary alloys that form BMGs, and since our work has demonstrated that ordering in the liquid influences both crystal nucleation and the glass transition, they are interesting systems to study supercooled liquids. Over the past year we have found experimental evidence for a sudden change in the liquid structures as the copper concentration is increased to above 45 at.%, which is near a projected metastable eutectic in the phase diagram. We also have found evidence that suggests a new type of liquid-liquid phase transition in the supercooled liquids. The pair distribution function for a Cu54Zr46 liquid shows a sudden change with temperature, to what appears to be two prominent types of short-range order, which become more prominent with increased supercooling. Evidence for a phase transition is also observed in the specific heat. Molecular dynamics calculations published by others have predicted the formation of two types of clusters, Cu- and Zr-centered, which stabilize the liquid. In a related Cu-Zr alloy, we find a temperature below which the Zr-Zr bonds show a negative thermal expansion coefficient and density measurements show an anomaly. These data suggest that we are possibly looking at a new type of liquid/liquid phase transition, which may be important for glass formation. Our results may be the first experimental evidence for this, providing new insight into the ordering that may underlie BMG formation in these alloys.

Studies of Zr2Ni liquids – High energy X-ray diffraction and ab initio molecular dynamics (MD) simulations have been made in collaboration with Alan Goldman, Matt Kramer and their group at Iowa State University. These results indicate that the liquid contains a large population of fragmented clusters with pentagonal and distorted pentagonal faces, but the number of fully developed icosahedral fragments is rare. Interestingly, in both cases, the ordering changes little over the 500 K of cooling. All metrics show that the nearest-neighbor atomic configurations of the most deeply supercooled simulated liquid (1173 K) differ topologically and chemically from those in the stable C16 compound, even though the partial pair distributions are similar. The most significant structural change upon decreasing the temperature from 1673 to 1173 K is an increase in the population of Zr in Ni-centered clusters. The structural differences between the liquid and the C16 increase the nucleation barrier, explaining glass formation in the rapidly quenched alloys.

Possible fragile to strong transition in supercooled liquid - We have noticed clear signatures of maxima in the specific heat near 800oK for a Zr62Cu20Ni8Al10 liquid and between 700 and 800 K for the Vitralloy 106a liquid. In 106a this is accompanied by a density inflection and correlates with a fragile to strong transition that has been inferred from viscosity measurements by R. Busch (University of Gottingen, Germany). We are collaborating with Busch and plan to submit a paper by the end of summer, 2009, describing these results.

Bibliography Type:	Description: (Last Updated: 05/20/2025)
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